

Gauge and Homological Structures in Quantum Error Correction



Junichi Haruna

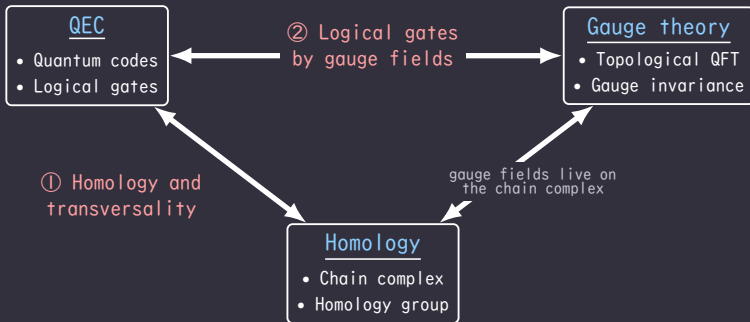
Kyoto University

Quantum computing seminar, University College London (Browne group)

9 Jul. 2026

Based on arXiv:2511.15224, 2602.14499

QEC – Homology – Gauge, a shared backbone



Claim 1. Whether a logical Z -rotation is transversal is decided by homology with $\mathbb{Z}_2$ coefficients. → Part II

Claim 2. Logical gates on any CSS code = exponentials of lattice gauge fields on its chain complex. → Part III

Talk plan

- Preliminary (quick recap) Part I
- Homological origin of transversal implementability (main result) Part II
- Logical gates by gauge field formalism Part III
- Summary Part IV

QEC = stabilizer codes

- Logical info $\hookrightarrow$ larger Hilbert space
- CSS code: commutative X/Z stabilizers

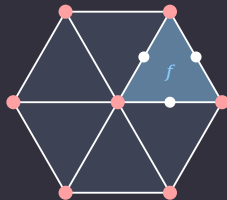
$$S_i^X = \prod_{j=1}^n (X_j)^{(H_X)_{ij}} \quad \text{or} \quad S_i^Z = \prod_{j=1}^n (Z_j)^{(H_Z)_{ij}}$$

- Stabilizers specify the structure of the code

CSS codes = chain complexes over $\mathbb{Z}_2$

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

- Lattice a space into cells
- Chain complex: $\partial =$ boundary map
- Qubits on C_1 ; $H_X = \partial_1$, $H_Z^T = \partial_2$
- CSS condition = $\partial^2 = 0$



face = C_2 , edge/qubit = C_1 , vertex = C_0

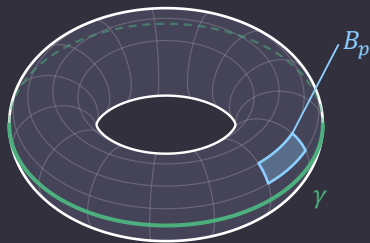
Logical operators = homology

- logical $Z = [\gamma]$

$$H_1 := \ker H_X / \text{im } H_Z^T$$

(Z -string mod stabilizers B_p)

- distance $d = \min$ weight of nontrivial cycle
- logical Pauli $\Rightarrow$ classified by **homology**



toric code

Chain complexes across QEC

- Measurement error $\Rightarrow$ longer chain complexes

- Fault-complex

[Hillmann et al. 2025]

- Single-shot QEC

[Campbell 2018; Jacob-McLauchlan-Browne 2508.08191]

- Algebraic operation $\Rightarrow$ new family of QEC codes

- Hypergraph product codes $\rightarrow$ tensor product

[Tillich, Zémor 2014]

- Fiber-bundle codes $\rightarrow$ fiber bundle

[Hastings, Haah, O'Donnell 2021]

- Lifted product codes $\rightarrow$ lifted product

[Panteleev, Kalachev 2021]

- Code surgery $\rightarrow$ mapping cone

[Ide et al. 2025]

Beyond Pauli $\rightarrow$ Clifford hierarchy

- Z : phase flip, X : bit flip $\Rightarrow$ not enough for universal quantum computation
- Universal gate set = climbing the Clifford hierarchy

$$\mathcal{P} = \mathcal{C}_1 \subset \mathcal{C}_2 \text{ (Clifford)} \subset \mathcal{C}_3 \subset \dots$$

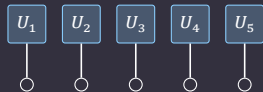
$$S = \begin{pmatrix} 1 & \\ & i \end{pmatrix} \in \mathcal{C}_2 \quad T = \begin{pmatrix} 1 & \\ & e^{i\pi/4} \end{pmatrix} \in \mathcal{C}_3$$

Problem

How to construct logical gates beyond Pauli in general?

Transversal gates, no unifying rule

- Transversal gate: automatically fault-tolerant
- Eastin-Knill: No transversal gate set
⇒ limits what is possible, not which gates for a given code
- Many transversal gates known, no unifying rule



no entangling gates

Problem 1

Which code can implement which gate transversally?

Two open problems

Problem 1

Which code can implement which gate transversally?

⇒ Part II [arXiv:2602.14499](#) (next)

Problem 2

How to construct logical gates beyond Pauli in general?

⇒ Part III [arXiv:2511.15224](#) (later)

Part II

Homological origin of transversal implementability

Junichi Haruna
arXiv:2602.14499

A zoo of touchstones, no principle

- Transversal gates $\Rightarrow$ `fault-tolerant`, but which gate for which code?
- Known touchstones for special codes:
 - Triorthogonal codes, 2^m -divisible codes, quantum Reed-Muller codes ...

Key questions

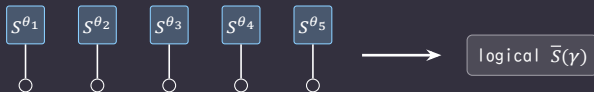
- When does a transversal gate give a logical gate?
- Is there `one principle` behind these touchstones?

Transversal Z-rotations

- Transversal Pauli Z-rotations:

$$U(\theta/2) := \prod_{j=1}^n \exp\left(i\pi \frac{\theta_j}{2} \frac{I - Z_j}{2}\right) = \prod_{j=1}^n (S_j)^{\theta_j} \quad \theta \in \mathbb{Z}_4^n$$

- Goal: when does a transversal S realize the logical $\bar{S}(\gamma) = \sqrt{\bar{Z}(\gamma)}$?



When is logical S transversal?

- Design test: hand me a code, can logical S be transversal?
- $S = \pi/2$ rotation of $Z \Rightarrow$ answer fixed by the stabilizers and $\bar{Z}$ alone

Result (preview)

- Transversal implementability of $S \Leftrightarrow$
two obstruction maps vanish for $\bar{Z}(\gamma)$; $\beta_1([\gamma]) = \beta_2([\gamma]) = 0$

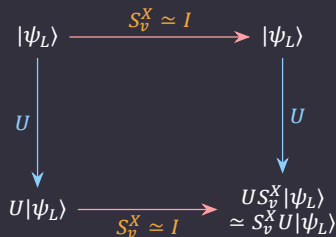
Logicality $\Leftrightarrow$ commuting with S^X

- Logicality $\Leftrightarrow$ commutativity on $\mathcal{H}_c$
- Commutation relation with X -stabilizer:

$$S^X(h)^\dagger U^\dagger \left(\frac{\theta}{2} \right) S^X(h) U \left(\frac{\theta}{2} \right) \\ = \underbrace{e^{i\pi \langle \theta, h \rangle / 2}}_{\text{const. phase}} \underbrace{U(\theta \circ h)}_{Z\text{-string}}$$

 $\theta \circ h$: entrywise product

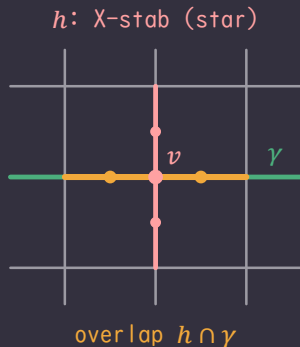
- Z-string must coincide with a stabilizer (and phase trivial)



require: $(S_v^X)^\dagger U^\dagger S_v^X U \simeq I$ on $\mathcal{H}_c$

Two conditions on θ , two moduli

- Two conditions on θ , different moduli
 - Phase factor: $H_X \theta = 0 \pmod{4}$
 - Z-string = stabilizer ($\perp h, \gamma$):
 $(H_X \odot K) \theta = 0 \pmod{2}$
- Which logical Z ? $\rightarrow$ further consideration ($S^2 = Z$)



cond. 1 pairs θ with h ; the Z -string $\theta \circ h$ lives on the amber overlap

K = stack of basis of $\ker H_7$; $\odot$: entrywise row products (cf. $\theta \circ h$)

Fixing the logical Z via $S^2=Z$

- Constraint: square of $U(\theta/2)$ must be $\bar{Z}(\gamma)$

$$U(\theta/2)^2 = \prod_i [(S_i)^{\theta_i}]^2 = \prod_i Z_i^{\theta_i} \stackrel{!}{=} \bar{Z}(\gamma)$$

- Additional constraint on θ with some $\mathbb{Z}_2$ vector ω

$$\theta = \gamma \pmod{2} \iff \exists \omega : \theta = \gamma + 2\omega \pmod{4}$$

Main theorem

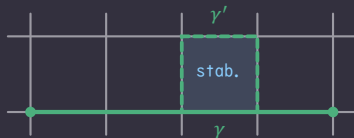
Theorem. Transversal implementability of logical S gate
 $\Leftrightarrow$ vanishing of two obstruction maps for $\bar{Z}(\gamma)$:

$$\beta_1([\gamma]) = \beta_2([\gamma]) = 0$$

- $\beta_1 := [(H_X \odot K)\gamma] \in \mathbb{Z}_2^n$: additional condition mod 2
- $\beta_2 := [H_X \gamma / 2] \in \mathbb{Z}_2^n / \text{im } H_X$: liftability condition modulo 4

β_1, β_2 live on homology

- β from *one* representative γ
- $\bar{Z}$: $\gamma \bmod$ stabilizers (many γ , one logical)
- Lemma: stabilizer shifts γ , not β
 $\Rightarrow \beta_1, \beta_2$ live on $\bar{Z}$, not γ

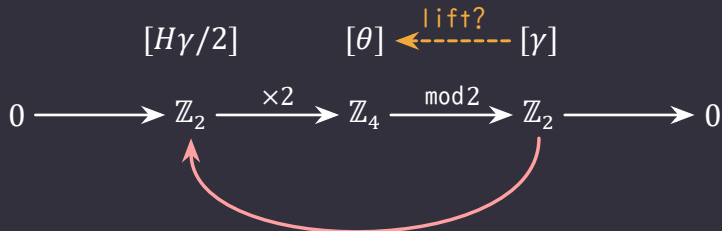


$$\gamma' = \gamma + \partial(\text{stab.})$$

$$[\gamma'] = [\gamma] \Rightarrow \beta(\gamma') = \beta(\gamma)$$

$\beta_2 =$ Bockstein obstruction

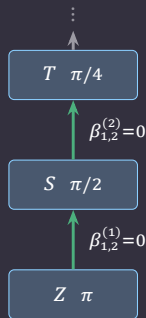
- Rotation angle $\pi \rightarrow \pi/2$ = lifting the coefficients $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$
- Transversal implementability = Existence of lift



lift exists $\iff \beta_2([\gamma]) = 0$ (keeping $[\gamma] = [\theta \bmod 2]$)

Climbing the Clifford hierarchy

- Same argument climbs the **Clifford hierarchy**: $Z \rightarrow S \rightarrow T \rightarrow \dots$
- Transversal $\Leftrightarrow$ a ladder of vanishing obstructions $\beta_1^{(v)} = \beta_2^{(v)} = 0$
- Known zoo = $\beta = 0$ cases
divisibility, tri-orthogonality



Steane $[[7, 1, 3]]$ (S yes, T no)

- Self-dual CSS ($H_X = H_Z = H$, the $[7, 4, 3]$ Hamming code): a single logical class (one γ to check: the table is the whole test)

gate	lift	obstruction	transversal?
$S(\pi/2)$	$1 \rightarrow 2$	$\beta_1^{(1)} = \beta_2^{(1)} = 0$	yes
$T(\pi/4)$	$2 \rightarrow 3$	$\beta_2^{(2)} \neq 0$	no

The known no-transversal- T on Steane recovered as $\beta_2^{(2)} \neq 0$: the obstruction detects it.

Summary of Part II

- Transversal logical $S \Leftrightarrow \beta_1([\gamma]) = \beta_2([\gamma]) = 0$ (two obstruction maps vanish)
- The β maps are **homology maps**; β_2 = Bockstein-type lifting obstruction

Transversal implementability of diagonal gates = a homological lifting obstruction
 $\Rightarrow$ from a blanket no-go to a computable obstruction theory (cf. Eastin-Knill).

Part III

Logical gates by gauge field formalism

Junichi Haruna
arXiv:2511.15224

Motivation

- Part II: an obstruction theory, but only for transversal `phase` gates
- Many codes, many logical gates, each construction `code-specific`
 - divisibility, tri-orthogonality, code automorphisms ...
- No general recipe, even with `fault tolerance set aside`

Key question

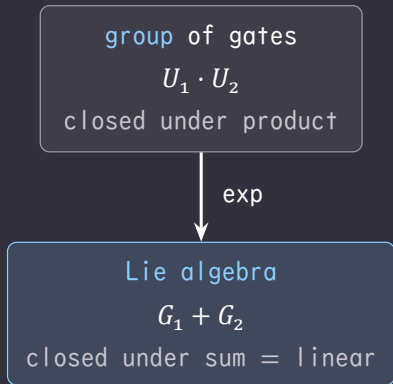
- How do we `build` a logical gate on an arbitrary CSS code?

Work in the exponent

- Gates form a **group**: closed under the product $U_1 U_2$ (nonlinear)
- Idea: push everything into the exponent

$$U = \exp(iG)$$

- Then generators just *add*: $G_1 + G_2$
⇒ a **linear space**, much simpler
- **Gauge field** = the generator G : an operator-valued cochain
linear map: qubits $\rightarrow$ operators



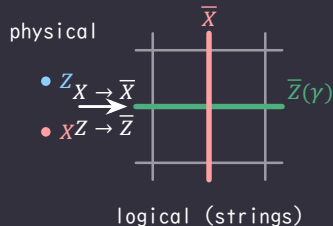
Replacement principle

- Any function of $\bar{X}, \bar{Z}$ is **logical**

$$[U(\bar{X}, \bar{Z}), S^X] = [U(\bar{X}, \bar{Z}), S^Z] = 0$$

- Which **gate**? $\rightarrow$ **Replacement principle**:
Replace physical X, Z with logical $\bar{X}, \bar{Z}$

$$\mathcal{O}(X, Z) \rightarrow \bar{\mathcal{O}}(\bar{X}, \bar{Z})$$



$\bar{\mathcal{O}}(\bar{X}, \bar{Z})$ commutes with every stabilizer $\Rightarrow$
logical

Gauge field = qubits $\rightarrow$ projectors

- Gauge field = linear map from qubit to operator
- Each qubit $e_j \rightarrow$ projectors onto $|1\rangle_j$:

$$a(e_j) := P_j^- = |1\rangle\langle 1|_j = \frac{I - Z_j}{2}$$

$$a(e_i + e_j) := a(e_i) + a(e_j)$$



linearity: $a(e_i + e_j) = a(e_i) + a(e_j)$

Why projectors?

- Every diagonal gate = one master formula:

$$U_{\text{diag}} = \exp\left(i f(P_1^-, \dots, P_n^-)\right)$$

- choosing the gate = choosing f :

$$Z = \exp(i\pi P^-)$$

same P^- ,

$$S = \exp\left(\frac{i\pi}{2} P^-\right)$$

only the angle changes

$$T = \exp\left(\frac{i\pi}{4} P^-\right)$$

$$CZ_{ij} = \exp\left(i\pi P_i^- P_j^-\right)$$

product $\rightarrow$ entangling

Building logical S ($n=3$)

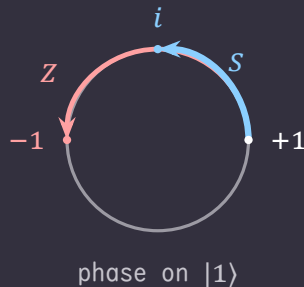
- S gate

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \exp \left(\frac{i\pi}{2} \frac{I - Z}{2} \right)$$

- Replacement principle: $Z \rightarrow \bar{Z}(\gamma)$ to get logical S

$$\bar{S}(\gamma) = \exp\left(\frac{i\pi}{2} \frac{I - \bar{Z}(\gamma)}{2}\right)$$

- How to decompose $\bar{Z}(\gamma)$ into physical gates?



Expanding $\bar{S}(\gamma)$ in projectors

- In **projectors**, from $\bar{Z}(\gamma) = Z_1 Z_2 Z_3$:

$$\bar{Z}(\gamma) = (I - 2P_1^-)(I - 2P_2^-)(I - 2P_3^-)$$

- Expanding (All terms are commutative)

$$\frac{I - \bar{Z}}{2} = P_1^- + P_2^- + P_3^- - 2(P_1^- P_2^- + P_1^- P_3^- + P_2^- P_3^-) + 4P_1^- P_2^- P_3^-$$

- The **cubic term = identity operator**

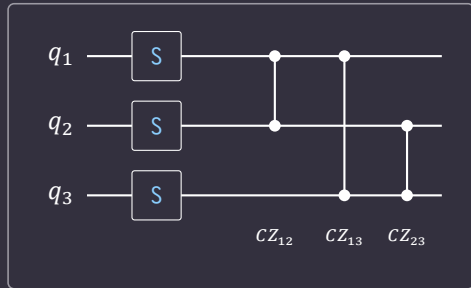
$$\exp(i\pi/2 \cdot 4P_1^- P_2^- P_3^-) = I$$

Physical circuit for $\bar{S}(\gamma)$

- Dropping the cubic term $\Rightarrow$ degree-2 $\mathbb{Z}_4$ phase polynomial

$$\begin{aligned} \overline{S}(\gamma) = & \underbrace{\exp\left(\frac{i\pi}{2}(P_1^- + P_2^- + P_3^-)\right)}_{S \text{ on each qubit}} \\ & \times \underbrace{\exp\left(i\pi(P_1^-P_2^- + P_1^-P_3^- + P_2^-P_3^-)\right)}_{CZ \text{ on every pair}} \end{aligned}$$

$$\bar{S}(\gamma) = S_1 S_2 S_3 \cdot CZ_{12} CZ_{13} CZ_{23}$$



From projectors to gauge fields

- Let $\gamma = e_1 + e_2 + e_3$,

$$a(\gamma) = P_1^- + P_2^- + P_3^-$$

$$a(\gamma)^2 = P_1^- + P_2^- + P_3^- + 2(P_1^- P_2^- + P_1^- P_3^- + P_2^- P_3^-)$$

- $\bar{S}(\gamma)$ in projectors

$$\bar{S}(\gamma) = \exp\left(i\frac{\pi}{2}(P_1^- + P_2^- + P_3^- + 2(P_1^- P_2^- + P_1^- P_3^- + P_2^- P_3^-))\right)$$

$$\bar{S}(\gamma) = \exp\left(i\frac{\pi}{2}a(\gamma)^2\right)$$

Full Clifford + T on general CSS

- Same recipe, full Clifford + non-Clifford T

Logical gate	Gauge field expression
$\bar{S}(\gamma)$	$\exp(i\pi a(\gamma)^2/2)$
$\bar{H}(\gamma)$	$\bar{S}(\gamma) \exp(i\pi b(\tilde{\gamma})^2/2) \bar{S}(\gamma)$
$\bar{CZ}(\gamma_1, \gamma_2)$	$\exp(i\pi a(\gamma_1)a(\gamma_2))$
$\bar{T}(\gamma)$	$\exp(i\pi (2a(\gamma)^3 - 3a(\gamma)^2 + 2a(\gamma))/4)$

- No special structure on QEC codes: applicable for general CSS codes

Two circuits for one $\overline{S}$

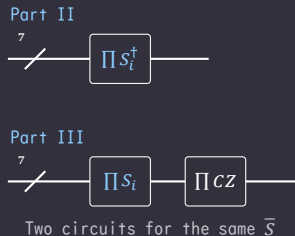
- Part II says

$$\bar{S} = \prod_{i=1}^7 S_i^{\dagger}$$

- Part III says

$$\bar{S} = \prod_{i=1}^7 S_i \cdot \prod_{1 \leq i < j \leq 7} CZ_{ij}$$

- Contradiction? What happens?



Resolved by a logical identity

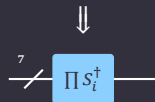
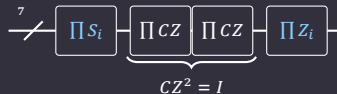
- CZ part + logical Z = logical identity

$$\prod_{1 \leq i < j \leq 7} CZ_{ij} \cdot \bar{Z} \simeq I$$

- Deform by a logical identity:

$$\prod_i S_i \cdot \prod_{i < j} CZ_{ij} \simeq \prod_i S_i \cdot \prod_i Z_i = \prod_i S_i^\dagger$$

- “Nice” logical gate $\Leftrightarrow$ “nice” logical identity



Deform $\bar{S}$ by a logical identity
 $\Rightarrow$ transversal $S^\dagger$

Summary of Part III

- **Dictionary**: logical gates in CSS codes $\Leftrightarrow$ polynomial functions of projectors (gauge fields in high energy physics)
- **Methods**: Replacement principle with functions of logical Pauli operators
- “Nice” logical gates $\Leftrightarrow$ “nice” logical identities

Gauge fields build the logical gate; **homology** decides if it is transversal
 $\Rightarrow$ two faces of **one structure**: the chain complex of the code.

Part IV

Summary and Future directions

Summary

Quantum error correction · homology theory · Gauge theory
share common mathematical structures.

- Logical gates: built systematically from lattice gauge fields
- Transversality: governed by homology beyond $\mathbb{Z}_2$
- Their interplay → new, unifying insight

Future directions

- **Fault-tolerance** written in the language of lattice gauge theory / homology.
- **Non-Abelian extension**: non-Abelian codes & the quantum double model (Kitaev).
- Higher cohomological invariants and non-Clifford gates.

A shared language, still mostly unexplored.

Thank you!

Backup

Steane $[[7, 1, 3]]$ computations

- S : lift $\hat{\theta} = 3 \cdot 1_7$ gives $\langle \hat{\theta}, 1_7 \rangle = 1 \pmod{4}$, so $\prod_i S_i^\dagger$ is logical S ($1_7 = \gamma$; realizes $\beta_1^{(1)} = \beta_2^{(1)} = 0$)
- T : every logical $\theta^{(3)} = 2c_0 1_7 + 4 \sum_i c_i h_i$ is even (h_i : rows of H), so $\langle \theta^{(3)}, 1_7 \rangle \neq 1 \pmod{8}$: no transversal T , even for non-uniform $\theta \in \mathbb{Z}_8^7$ ($= \beta_2^{(2)} \neq 0$ in the table)

Recovering known criteria

The framework reproduces the known sufficient conditions:

Known criterion	In this framework
Divisibility: a 2^m -divisible CSS code admits transversal $\exp(i\pi Z/2^m)$	$\beta_2^{(v)} = 0$ for uniform θ_v (the homology class lifts)
Tri-orthogonality $\Rightarrow$ transversal T	level $m = 3$ with $\beta_1^{(v)} = \beta_2^{(v)} = 0$ (uniform angle)
Quasi- / multi-level generalizations	special non-uniform θ_v (still sufficient)

v : one obstruction pair per lift step $\mathbb{Z}_2^v \rightarrow \mathbb{Z}_2^{v+1}$

Divisibility and tri-orthogonality are the uniform-angle corner ($\beta_{1,2}^{(v)} = 0$) of a full homological classification; non-uniform θ_v give **strictly more** transversal gates (examples: arXiv:2602.14499).

Tri-orthogonality is also the Bravyi-Haah criterion for low-overhead magic-state distillation.

Logical operators = gauge invariant

- Logical U preserves the code space:

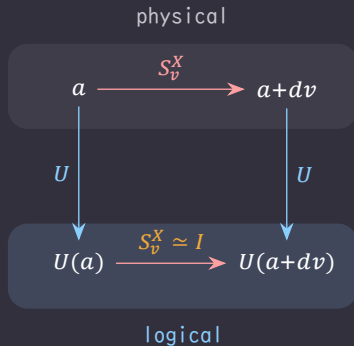
$$g^\dagger U g \simeq U \text{ on } \mathcal{H}_c \quad (g: \text{any stabilizer})$$

- Conjugating by an X -stabilizer sends $a \rightarrow a + dv$:

$$(S_v^X)^\dagger a S_v^X = a + dv \pmod{2}$$

- Logical = **gauge-invariant**: any $U(a)$ built from the field depends only on $[a] \in H^1$
($H^1 \cong H_1$ over $\mathbb{F}_2$: same classes as Part I)

$$\begin{aligned} (S_v^X)^\dagger U(a) S_v^X &= U(a + dv) \\ &\simeq U(a) \end{aligned}$$

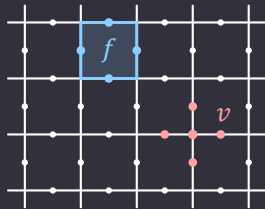


Gauge field formalism (imported)

Where do the β maps come from?

Logical operators = exp(polynomial in gauge fields): an imported tool.

Quantum error correction	Gauge theory (import)
X-stabilizer S^X	Gauss-law generator
Z-stabilizer S^Z	Magnetic flux
Code space $\mathcal{H}_c$	(Gauge-invariant) ground states
Logical $\bar{Z}(\gamma)$	Wilson loop $W(\gamma)$



Imported from $\mathbb{Z}_2$ lattice gauge theory (Kogut-Susskind):

$$H = - \sum_v A_v - \sum_p B_p$$

(= the toric code Hamiltonian)

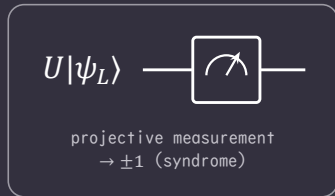
$$\text{Wilson loop } W(\gamma) = \prod_{e \in \gamma} Z_e.$$

Detecting and correcting errors

- Measure stabilizers $\Rightarrow$ syndrome, decode & correct

$$|\psi_L\rangle \xrightarrow{\text{error}} U|\psi_L\rangle \xrightarrow{\text{measure}} S \rightarrow \pm 1$$

- Code distance $d = \min.$ weight logical error
(undetected, but acts nontrivially)



Relation to Webster-Quintavalle-Bartlett

Their XP-stabiliser algorithm (arXiv:2303.15615) classifies transversal diagonal logicals of any stabiliser code.

At each level m , the two classifications coincide: $V_{LD}^{(m)} = V_L^{(m)} / V_{LI}^{(m)}$ (conditions match via inclusion-exclusion mod 2).

	WQB (2303.15615)	This work
Method	implicit kernels, per-code algorithm (Howell / Smith normal form over $\mathbb{Z}_2^m$)	a closed form for an arbitrary CSS code
Output	a generating set, code by code	obstruction maps β_1, β_2 on homology
Scope	multi-qubit controlled-phase; non-CSS via embedding	inter-level lifting; absence seen beyond uniform angle

Same classification, complementary content: WQB compute it code by code; the homological form exposes the structure and carries the lifting obstruction (β_2 Bockstein-type).

[[15, 1, 3]] Reed-Muller setup

Sanity check: does the homological criterion reproduce a result you already know?

- Setup: the [[15, 1, 3]] punctured Reed-Muller code. Physical $T^\dagger$ on every qubit $\rightarrow$ logical T .
- In this framework: uniform angle $\theta_i \equiv 1$, level $m = 3$, $\mathbb{Z}_8$ coefficients.
- **Tri-orthogonality** (every triple overlap of H^Z rows is even; = the magic-state distillation condition) $\Rightarrow \beta_1^{(\nu)}, \beta_2^{(\nu)}$ **vanish** for $\nu = 1, 2, 3$.

Re-deriving a known result: the textbook **tri-orthogonal** $\Rightarrow$ **transversal** T is about to drop out of the same two obstructions, now at uniform angle, level 3.

[[15, 1, 3]] transversal T

- Both obstructions vanish $\Rightarrow$ a $\pi/4$ rotation of logical Z is transversal = transversal T .
- Gauge reading: $m = 3$, so logical $\bar{T}$ is the $2^{m-1} = 4$ -th root of the Wilson loop (imported gauge object):

$$\bar{T}(\gamma) = W(\gamma)^{1/4}$$

Textbook tri-orthogonal $\Rightarrow$ transversal T = both obstructions vanish at uniform angle, level 3.

Recovering gates from gauge fields

Imported tool: $a(e)$ electric, $b(e)$ magnetic gauge field on edge e .

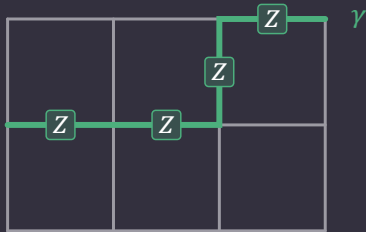
- Single-qubit gates come straight back:

$$Z_j = \exp(i\pi a(e_j)),$$

$$X_j = \exp(i\pi b(e_j))$$

- Transversal Z gates =
gauge-field exponentials:

$$\prod_{i=1}^n (Z_i)^{\gamma_i} = \exp(i\pi a(\gamma))$$



Stabilizers & logical operators

- Stabilizers from the (co)boundary maps:

$$S_f^Z = \exp(i\pi a(\partial f)),$$

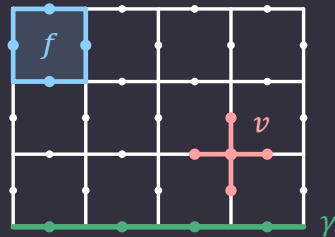
$$S_v^X = \exp(i\pi b(\partial^T v))$$

- Logical Z operator = Wilson loop:

$$\bar{Z}(\gamma) = \exp(i\pi a(\gamma))$$

$$\leftrightarrow W(\gamma) = \exp\left(i \oint_{\gamma} a\right)$$

($\oint_{\gamma} a$: continuum notation for $a(\gamma)$)



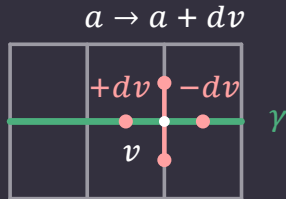
Construction from gauge invariance

- Any function $U(a(\gamma))$ is gauge invariant:

$$a(\gamma) = \oint_{\gamma} a \rightarrow \oint_{\gamma} (a + dv) = \oint_{\gamma} a$$

- Which logical gate does it implement? $\rightarrow$
Replacement principle
- Replace physical X, Z with logical $\bar{X}, \bar{Z}$:

$$\mathcal{O}(X, Z) \rightarrow \bar{\mathcal{O}}(\bar{X}, \bar{Z})$$



$+dv - dv = 0$ along $\gamma \Rightarrow \oint_{\gamma}$ unchanged

Cohomology dependence

- Logical action depends only on the **cohomology class**
- Gauge transformation $a(\gamma) \rightarrow a(\gamma) + dv$: Logical operator = **Gauge-invariant** operator

$$U(\gamma) \simeq U(\gamma + \partial_2 f) \quad \text{on } \mathcal{H}_c$$

Open questions:

- Incorporating **fault-tolerance** or **circuit-depth** optimality
- Higher-order topological invariants; higher-form gauge fields

Level- m transversal gate = 2^{m-1} -th root

- Transversal Pauli Z rotation at level m = a 2^{m-1} -th root of the logical $Z(\gamma)$
- Implementability of the transversal gate = existence of that (gauge-invariant) root
- Example: $m = 2$ = transversal S (our running case)

gauge field side (imported)

$$W(\gamma) = \exp\left(\oint_{\gamma} a\right)$$

$\leftrightarrow$

QEC operator side

$$Z(\gamma) = \prod_i (Z_i)^{\gamma_i}$$

$$W^{(2)}(\gamma^{(2)}) = \exp\left(\frac{1}{2} \oint_{\gamma^{(2)}} \tilde{a}\right)$$

$\leftrightarrow$

$$S(\gamma^{(2)}) = \prod_{i=1}^n (S_i)^{\gamma_i^{(2)}}$$

$$\text{with } W(\gamma) = (W^{(2)}(\gamma^{(2)}))^2$$

$\tilde{a}$, $\gamma^{(2)}$: the level-2 field and chain from the logical- S construction

Our transversal S is the case $m = 2$: the **square root** of the Wilson loop.
 $m = 3$ (the 4th root) = transversal T on $[[15,1,3]] \rightarrow$ the $[[15,1,3]]$ backup slides.

Non-Abelian code and quantum double

- Non-Abelian gauge groups $\rightarrow$ non-Abelian anyons
- Quantum double model $D(G)$ (Kitaev 1997): a lattice code for any finite group G ; $G = \mathbb{Z}_2$ = toric code, non-Abelian G (e.g. S_3, D_4) $\rightarrow$ non-Abelian anyons
- Open question: does the $\mathbb{Z}_2$ obstruction framework extend to non-Abelian G ? ($\rightarrow$ next)

Non-Abelian anyons and universal computation

- Non-Abelian anyons: matrix-valued exchange statistics $\Rightarrow$ universal topological QC
- Non-Abelian CSS? CSS is $\mathbb{Z}_2$ -based $\rightarrow$ generalization is a hot topic

Extending the gauge-field / homology dictionary ($a(e), b(e)$; the β obstructions) from $\mathbb{Z}_2$ to non-Abelian G is wide open, and these tools should transfer.